# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — SOCIAL SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): Climate and weather are the same. Reason (R): Both describe the state of the atmosphere. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** Which lists the layers of the atmosphere in order from the surface upward? 1

**(a)** Stratosphere, troposphere, mesosphere, exosphere, thermosphere
**(b)** Troposphere, mesosphere, stratosphere, exosphere, thermosphere
**(c)** Troposphere, stratosphere, mesosphere, thermosphere, exosphere
**(d)** Exosphere, thermosphere, mesosphere, troposphere, stratosphere

**3.** If a place has 30 years of data showing 20 days rain in July each year, this is part of its 1

**(a)** season **(b)** pressure **(c)** climate **(d)** weather

**4.** Which is NOT a source of greenhouse gases? 1

**(a)** Deforestation **(b)** Solar panels **(c)** Burning coal **(d)** Cattle farming

**5.** Assertion (A): Aeroplanes fly in the stratosphere. Reason (R): It is calm and free from clouds. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**6.** Nakshatras are traditionally used to 1

**(a)** track aurora **(b)** measure rainfall
**(c)** find the ozone layer **(d)** divide the Moon's path into 27 parts

**7.** Which statement correctly explains why the temperature falls from equator to poles? 1

**(a)** Humidity increases **(b)** Insolation decreases
**(c)** Rain increases **(d)** Pressure increases

**8.** The reason why we do not feel air pressure is 1

**(a)** gravity absent **(b)** internal pressure equals external
**(c)** no air presses **(d)** air is light

**9.** **Assertion (A):** The north-east monsoon brings rain to the Tamil Nadu coast. 1
**Reason (R):** The winds pick up moisture over the Bay of Bengal before reaching the coast.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** Nitrogen is the most abundant gas in the atmosphere. 1
**Reason (R):** Oxygen forms about 21 per cent of the atmosphere.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Explain why oceans reduce the impact of climate change on temperature but not on sea level. 2

**12.** Why can't a single hot day prove climate change? 2

**13.** What does 'climate smart' mean in Mission Mausam? 2

**14.** Why is the atmosphere compared to a blanket? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** What can individuals and communities do after floods to reduce disease? 3

**16.** Analyse how forests and fossil fuels affect the carbon footprint. 3

**17.** Explain with an example how humidity and temperature together affect comfort. 3

OR

Why does the NE monsoon rain Tamil Nadu but bring cold dry winds to Delhi? 3

**18.** A school wants to cut its carbon footprint. Suggest three actions. 3

OR

Explain why the troposphere is thicker at the equator than at the poles. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** During the monsoon of 2025, the rivers Satluj, Beas and Ravi swelled and flooded large parts of Punjab. Weak embankments and construction near rivers worsened the damage. Scientists link rising carbon dioxide and methane in the air with global warming, and advise planting trees, cycling and using solar energy. 5
(i) Name two rivers that flooded Punjab in 2025.
(ii) In which season did these floods occur?
(iii) Name one greenhouse gas.
(iv) Give one human factor that worsened the floods and one action that reduces our carbon footprint.

**20.** Analyse the atmosphere's role as a blanket protecting life, with reference to the layers. 5

OR

Evaluate whether individual action is enough to tackle climate change. 5

## END OF QUESTION PAPER — Total: 40 Marks